

GenAI Brainstorming & Model Selection

Foundations of AI Workshop Series · Workshop 3

Mashrekur Rahman, PhD

Research Facilitation, Dartmouth Libraries

Winter 2026

Today You'll Learn

- How LLMs reason & generate text
- Navigate Dartmouth Chat's models
- Tune prompts & parameters effectively
- Match the right model to your task

Quick Recap: Workshops 1 & 2

Building Your AI Foundation

Workshop 1: NLP to LLMs



AI Learning Loop

Training cycles: predict → compare → adjust



Text → Numbers

Encodings & embeddings capture meaning



Attention Mechanism

Words "attend" to relevant context



Transformers

Foundation of modern LLMs

Workshop 2: Deep Learning & Vision



Neural Networks

Layers of neurons learn patterns



CNNs (Spatial)

Filters detect edges → shapes → objects



LSTMs (Temporal)

Memory gates: forget, input, output



Diffusion Models

Noise → image through denoising

Today's Workshop Agenda



10 min

How LLMs Think

Token prediction, reasoning, and model evolution



20 min

Model Selection

DartmouthChat, models, comparisons, and Pareto tradeoffs, activity 1 & 2



14 min

Prompting Techniques

System prompts, parameters, and prompt engineering, activity 3



15 min

Use Case Exploration

Think-Pair-Share: matching models to real workflows

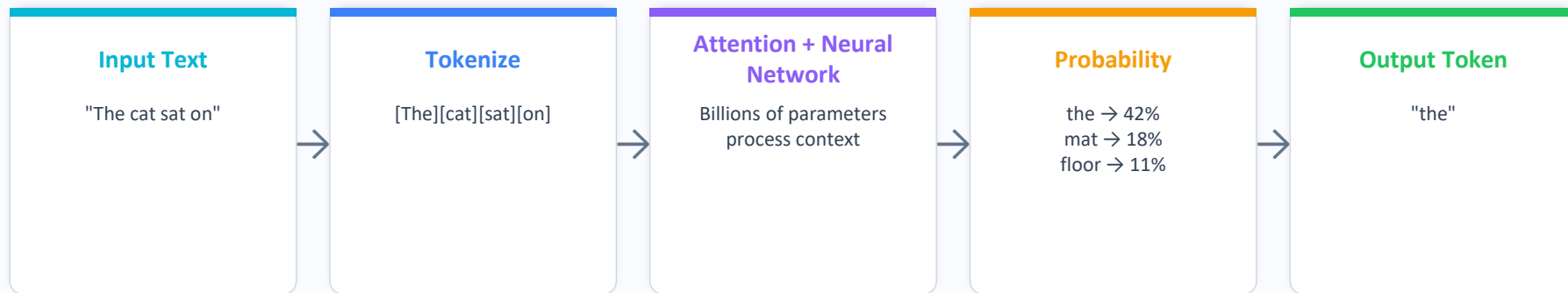


1 min

Wrap-Up & Resources

How LLMs Work – Connect to W1

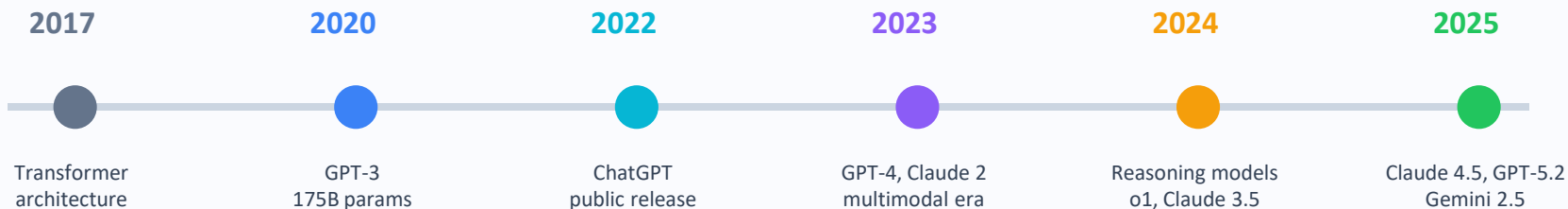
Next-token prediction



Key Insight

LLMs don't "understand"! They predict from a plausible space. One token at a time, the model picks the most likely next word based on patterns learned from trillions of examples.

The Rapid Evolution of LLMs



Size → Efficiency

Smaller models now rival earlier giants

Text → Multimodal

Images, audio, video alongside text

Generation → Reasoning

Chain-of-thought and structured thinking

How Models Reason

Different architectures produce different thinking styles



Standard Generation

How it works

Single forward pass — fast

Best for

Simple Q&A, summarization, translation

DartmouthChat models

Haiku 4.5, GPT-5.2 Instant, Gemini Flash

Fastest



Chain-of-Thought

How it works

"Think step by step" in prompt

Best for

Math, logic, multi-step reasoning

DartmouthChat models

Sonnet 4.5, Gemini 2.5 Pro, Mistral Large 3

Moderate



Built-in Reasoning

How it works

Extended internal "thinking" phase

Best for

Complex analysis, coding, research

DartmouthChat models

Opus 4.5, GPT-5.2 Thinking

Slower but deeper

Activity 1: Models on chat.Dartmouth.edu

chat.dartmouth.edu — 13 models across 5 providers

Cloud / Enterprise APIs

Claude Haiku 4.5	Fast & affordable
Claude Sonnet 4.5	Balanced workhorse
Claude Opus 4.5	Most capable
GPT-5.2 Instant	Fast generation
GPT-5.2 Thinking	Extended reasoning
Gemini 2.0 Flash	Ultra-fast
Gemini 2.5 Flash	Fast + smart
Gemini 2.5 Pro	Premium reasoning
Mistral Large 3	Strong multilingual

Open Source / On-Premise

GPT-OSS 120B

Open-source variant, large capacity

Gemma 3 27B

Google's efficient open model

Llama 3.2 11B

Meta's lightweight model

Qwen3-VL 32B

Vision + language multimodal

Open-source: runs on Dartmouth infrastructure

Model Families: Size vs. Capability

Within each provider, bigger models are slower and costlier — but more capable

Anthropic — Claude

Haiku 4.5

Speed ●●●●●

Quality ●●●●○

Fast

Sonnet 4.5

Speed ●●●●○

Quality ●●●●○

Balanced

Opus 4.5

Speed ●●○○○

Quality ●●●●●

Best

OpenAI — GPT

GPT-5.2 Instant

Speed ●●●●○

Quality ●●●●○

Fast

GPT-5.2 Thinking

Speed ●●○○○

Quality ●●●●●

Deep

Google — Gemini

2.0 Flash

Speed ●●●●●

Quality ●●●●○

Fastest

2.5 Flash

Speed ●●●●○

Quality ●●●●○

Smart

2.5 Pro

Speed ●●○○○

Quality ●●●●●

Top

The Tradeoff Principle

Every model family offers a spectrum from fast/cheap to slow/powerful. The right choice depends on your task:
simple tasks → use the smallest model | complex research → use the most capable

Efficiency & Resource Tradeoffs

Larger models need more compute, energy, and time — choosing wisely saves energy

Model	Parameters	Rel. Speed	Rel. Cost	Energy/Query	Best For
Llama 3.2 11B	11 billion	●●●●●	Free*	~0.5 Wh	Quick lookups, drafts
Gemma 3 27B	27 billion	●●●●○	Free*	~1.2 Wh	Summaries, analysis
Qwen3-VL 32B	32 billion	●●●○○	Free*	~1.5 Wh	Vision + text tasks
Claude Haiku 4.5	~20B est.	●●●●●	\$	~0.8 Wh	Fast classification
Claude Sonnet 4.5	~70B est.	●●●○○	\$\$	~3 Wh	Writing, coding, analysis
Claude Opus 4.5	~200B est.	●●○○○	\$\$\$	~8 Wh	Complex reasoning
GPT-5.2 Thinking	~500B est.	●○○○○	\$\$\$\$	~12 Wh	Hardest problems only

* Open-source models run on Dartmouth infrastructure

80% of tasks

Can be handled by smaller, faster models at a fraction of the cost and energy.

When you need the best

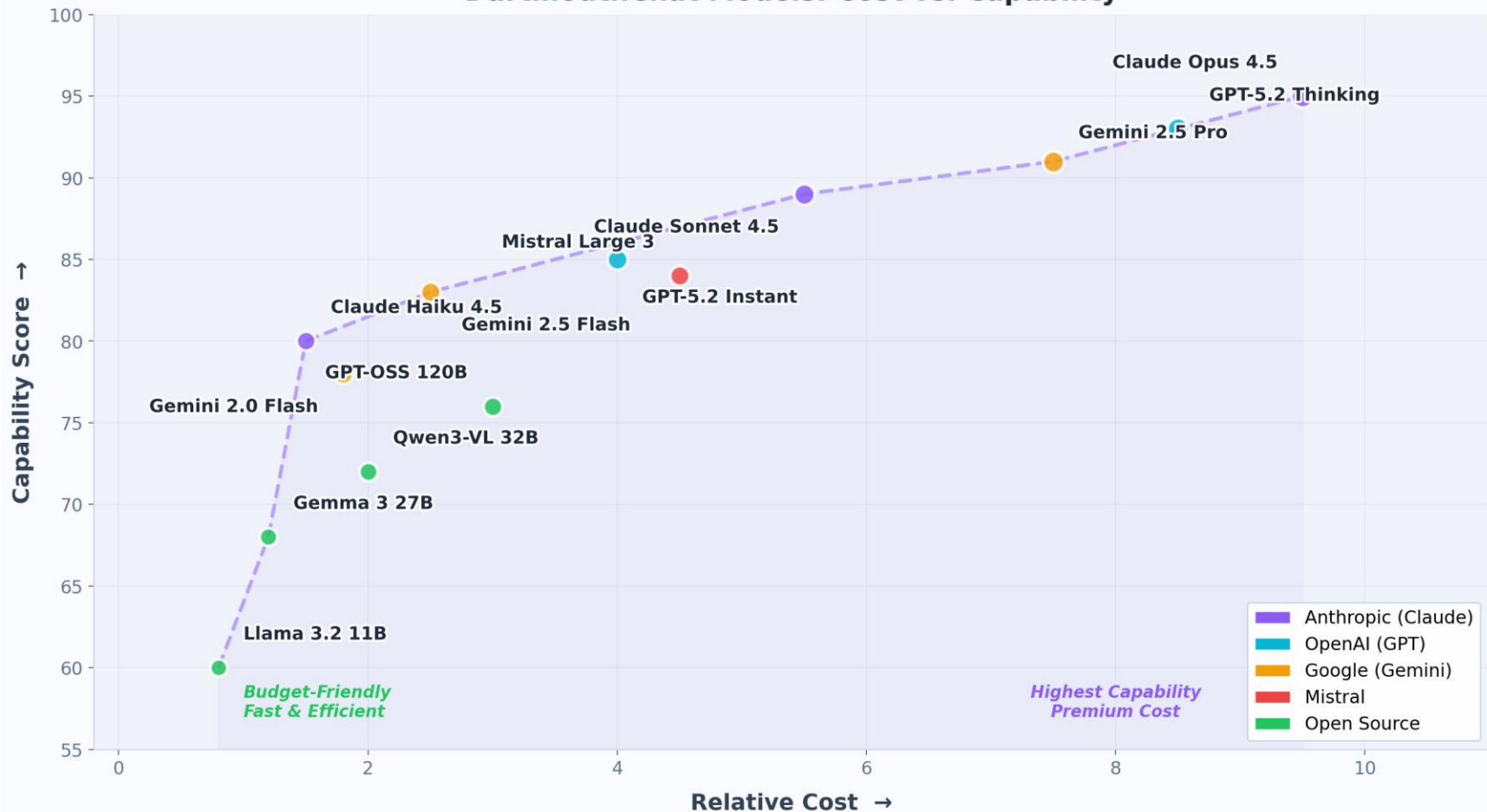
Reserve Opus, GPT-5.2 Thinking, or Gemini Pro for tasks where quality clearly matters.

Activity 2:

The Pareto Frontier: Cost vs. Capability

Fill the chart with the 13 models available through Dartmouth Chat

DartmouthChat Models: Cost vs. Capability



No single model is best for everything! Match the model to your task, budget, and quality needs

Prompt Engineering Essentials

Be Specific

"Write about dogs"
→ "Write a 200-word paragraph about the history of golden retrievers as family pets"

Give Examples

"Format like this:"

- Finding: [what you found]
- Confidence: [high/medium/low]
- Source: [where from]"

Set Constraints

"3 bullet points maximum.
Language a 10-year-old understands.
No technical jargon."

Chain Tasks

"First, identify the main argument.
Then, list 3 counter-arguments.
Finally, evaluate which is strongest."

System Prompts: Setting the Stage

The system prompt shapes everything the LLM does — your backstage instructions

Anatomy of a System Prompt

Role

You are a research librarian specializing in digital humanities.

Context

You help Dartmouth faculty find and evaluate academic sources.

Format

Respond with numbered lists. Cite sources in APA format.

Constraints

Only recommend peer-reviewed sources. Say "unsure" when uncertain.

On DartmouthChat

In the Chat Controls panel, find the System Prompt field at the top.

Paste your instructions before starting the conversation.

Pro Tips

- Be specific about format & constraints
- Include examples of desired output
- System prompt persists across the chat
- Combine with Function Calling for tools
- Use Reasoning Tags to see thinking

Activity 3: System Prompt Lab

Three real-world system prompts you can try on DartmouthChat

Research Assistant

You are a meticulous research assistant.

When given a topic, provide:

1. Brief overview (3-4 sentences)
2. Key findings from recent literature
3. Gaps or open questions

Always cite sources. If unsure, say so.

Code Reviewer

You are a senior software engineer.

Review code for:

1. Bugs and edge cases
2. Performance issues
3. Readability improvements

Use a friendly but direct tone.
Show corrected code.

Teaching Assistant

You are a patient tutor.

When a student asks a question:

1. Don't give the answer directly
2. Ask a guiding question instead
3. Build on what they already know

Adjust language to their level.

Your Model Selection Checklist

Question	Simpler Option	More Capable Option
How complex is the task?	Simple → Haiku, Flash, Llama	Complex → Opus, GPT-5.2 Thinking, Pro
How fast do you need it?	Real-time → Flash models, Haiku	Can wait → Opus, GPT-5.2 Thinking
Data privacy?	Standard → Any cloud model	Sensitive → Open-source (GPT-OSS, Llama, Gemma)
What output format?	Free text → Standard generation	Structured → System prompt + format controls
Do you need reasoning?	Recall/summary → Standard models	Analysis/math → Thinking, Opus

Activity 4: Use Case Exploration

Think → Pair → Share

3 min think · 3 min pair · 4 min share

Think

What task in your work is repetitive or time-consuming?

What would the ideal AI output look like?

What constraints matter? (privacy, accuracy, speed)

Pair

Share your use case with a neighbor

Help each other pick a model from DartmouthChat

Discuss

Share

Volunteer to share with the group

What surprised you about model differences?

What would you try next?

Activity 4: Scenarios

Think → Pair → Share

3 min think · 3 min pair · 4 min share

Scenario A: Research Assistant

Task:

Summarize 20 papers on climate science, extract key findings and contradictions

Discussion:

Which model? How to structure the prompt?

Scenario B: Course Material

Task:

Generate quiz questions from a textbook chapter, with distractors and explanations

Discussion:

Speed vs. accuracy tradeoff? Which reasoning level?

Scenario C: Data Analysis

Task:

Analyze 500 free-text survey responses to identify themes and sentiment

Discussion:

Can a small model do this? Context length? Privacy?

Key Takeaways



LLMs predict one token at a time — this simple mechanism produces emergent reasoning and creativity.



DartmouthChat gives you 13 models — start with fast/cheap models and scale up only when needed.



System prompts + parameters let you customize behavior without any coding — experiment freely!



80% of tasks work great with smaller models. Save premium models for harder problems.



Open-source models on Dartmouth infrastructure keep your data on campus — ideal for sensitive work.

Thank You!

Keep exploring GenAI

Resources

- DartmouthChat → chat.dartmouth.edu
- Follow-up Questions → mashrekur.rahman@dartmouth.edu

Workshop Series

W1: NLP to LLMs

W2: Deep Learning & Computer Vision

W3: GenAI Brainstorming ← You are here

W4: Critical AI Literacy → Coming next!